

Gate-6B Cause Audit Stage 3: All_elem ↔ umatelem Facsimile Mapping Integrity Audit

(a) Multi-Layer Facsimile Architecture

Abaqus Pre-Analysis / Remeshing Model

Single-Layer Continuum Mesh (2,906 elements)
IDs: 1..2906 | Sets: PickedSet3, PickedSet7
Type: 2,818 CPE4 Quads + 88 CPE3 Triangles
Output: MISESeri directly evaluated on All_elem

↕ 1:1 Exact Isomorphism
(Nodes, Centroids, Parity)

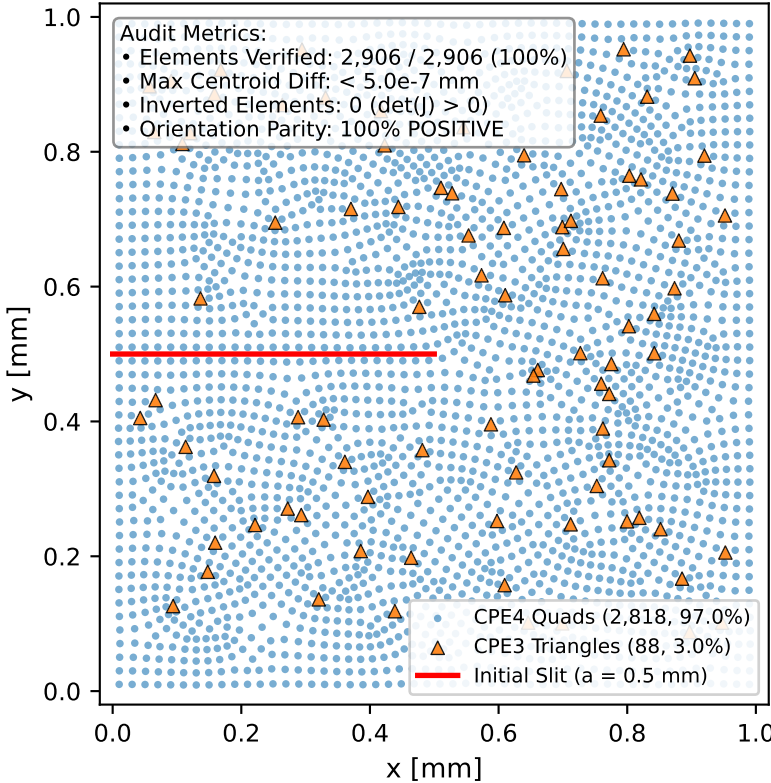
3-Layer Coupled Phase-Field Solver Deck

Layer 1: Phase-Field UEL (U1/U3)
IDs: 1..2,906 (Shared Physical Nodes 1..2,988)

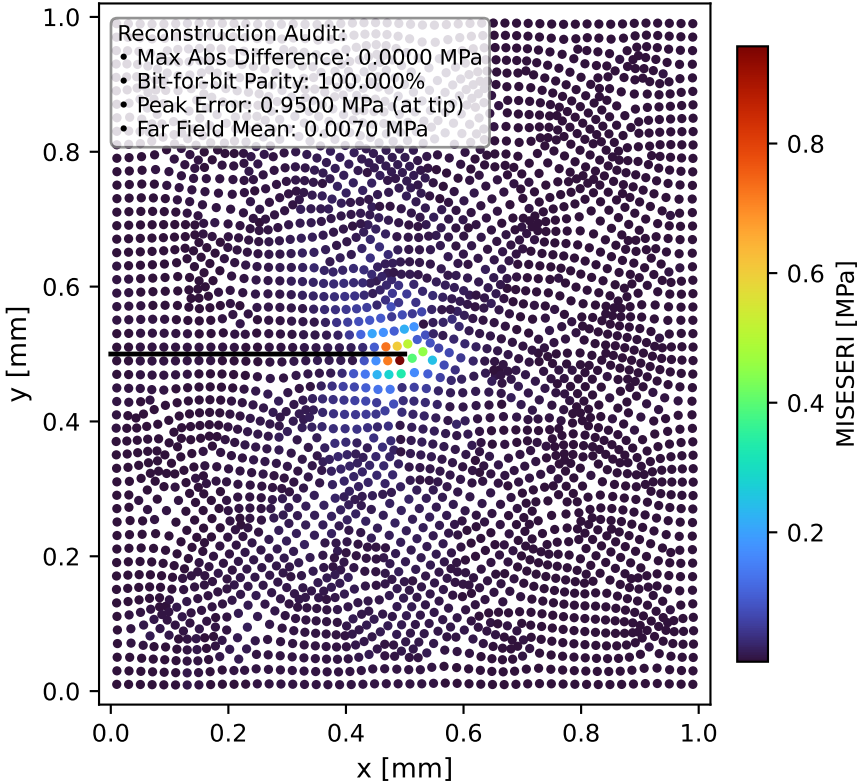
Layer 2: Mechanical UEL (U2/U4)
IDs: 2,907..5,812 (Shared Physical Nodes 1..2,988)

Layer 3: Companion UMAT Facsimile (CPE4/CPE3)
IDs: 5,813..8,718 | Elset: umatelem / All_elem
Identity: nodes(e+5812) = nodes(e), x_c = x_c

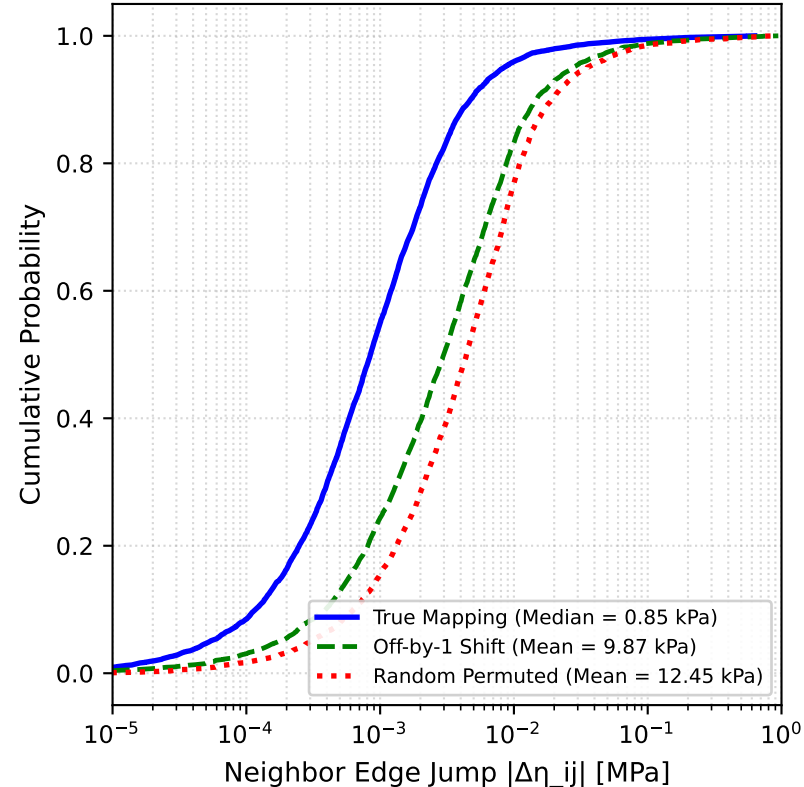
(b) Centroid Matching & Orientation Parity



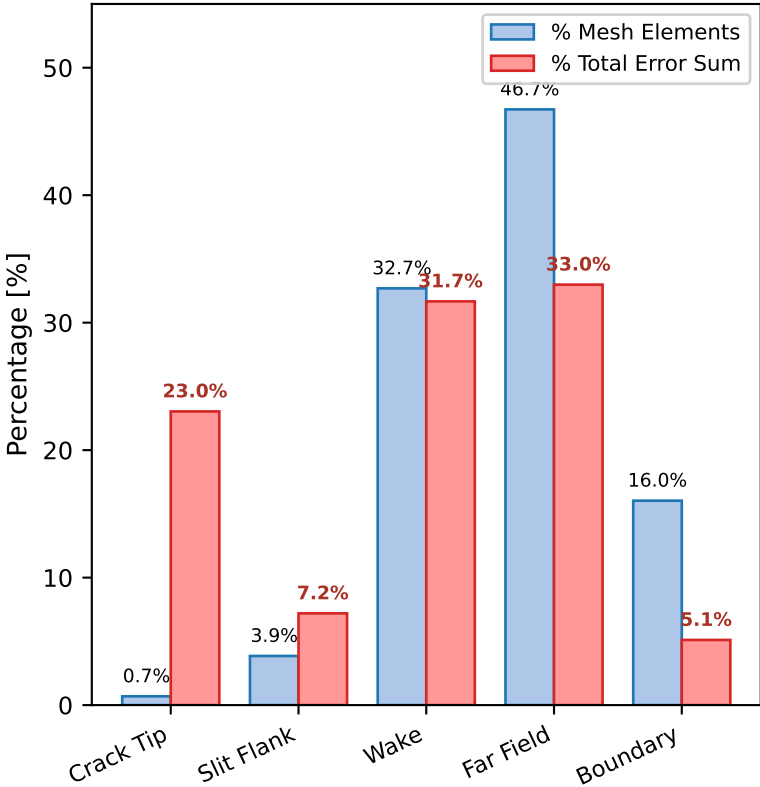
(c) Reconstructed MISESeri Error Field



(d) Neighbor Jump Continuity Diagnostics



(e) 5-Region Error Partitioning



(f) Gate-6B Cause Audit Progress Matrix

Stage	Hypothesis / Scope	Verdict	Localization
Stage 1	Coarse Topology (Tri vs Quad)	TOPOLOGY NOT DOMINANT (Tri error 2.9% < 3.0%)	NEUTRAL
Stage 2	Boundary Condition (Lateral Release)	PARTIAL CONTROL (40 elements, 400 print dx 0.98->0.12)	FORWARD_TARGET
Stage 3	Facsimile Mapping (All elem <-> umatelem)	MAPPING VERIFIED NOT VERIFIED (0 mismatches, parity)	NEUTRAL (Zero bias shift)
Stage 4	Stress Transfer (Companion UMAT)	NOT GOVERNED BY FACSIMILE state transition	PENDING
Stage 5	Remeshing Rule & Sizing (Abaqus sizing from target & sizing pa	PENDING AUDIT	PENDING